STPM 2023 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ *J s,* $c = 3.00 \times 10^{8}$ *m* s^{-1}*,* $1\ u = 1.66 \times 10^{-27}$ *kg.*

Section A [15 marks]

1. Which statement is true about simple harmonic motion?

A The acceleration is uniform.
B The amplitude decreases with time.
C The restoring force dissipates naturally.
D The kinetic and potential energies interchange continuously.

Answer: D

In simple harmonic motion the kinetic energy and potential energy continuously change into each other while the total energy stays constant.

Explanation: The acceleration is not uniform (it is proportional to displacement), and in SHM the amplitude stays constant (no damping).

2. A particle that undergoes simple harmonic motion has period of 2.0 s and amplitude of 2.4 cm. What is the velocity of the particle when the displacement is 1.2 cm?

A 6.5×10^{-2} m s^{-1} B 4.2×10^{-2} m s^{-1} C 6.6×10^{-3} m s^{-1} D 1.4×10^{-3} m s^{-1}

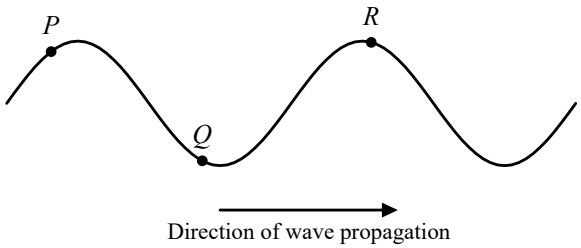
Answer: A

$$\omega = \frac{2\pi}{2.0} = \pi \text{ rad s}^{-1}$$

$$v = \omega\sqrt{(A^2 - x^2)} = \pi\sqrt{(0.024^2 - 0.012^2)} = 6.5 \times 10^{-2} \text{ m s}^{-1}$$

Explanation: Convert centimetres to metres. At half the amplitude the speed is $\frac{\sqrt{3}}{2}$ of the maximum speed.

3. A static picture of a travelling wave in a string with three particles, P, Q and R, is shown in the diagram below.



Which is the correct direction of motion for P, Q and R?

	P	Q	R
A	→	←	→
B	→	→	→
C	↓	↑	↑
D	↑	↓	↓

Answer: C

For a wave moving to the right, each particle copies the motion of the particle just to its left (behind it).

P (rising slope before a crest) moves down; Q (just before the trough's lowest point is passed) moves up; R (just past the crest) moves up: ↓ ↑ ↑.

Explanation: Particles in a transverse wave move perpendicular to the direction of propagation, never along it (A and B are wrong). Imagine the wave shape shifted slightly to the right to see which way each particle moves.

4. A man stands at a distance from a loudspeaker which emits sound uniformly in all directions. The sound energy received by the eardrums of the man for every second is 3.0×10^{-12} J. If the effective area of the eardrum is 80 mm^2, what is the intensity level of the sound at the man's eardrum? [Threshold intensity is 10^{-12} W m^{-2}.]

A 3.8 dB B 16 dB C 36 dB D 46 dB

Answer: D

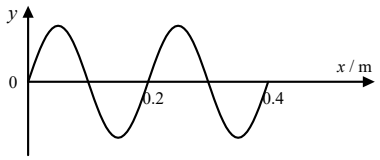
$$I = \frac{P}{A} = \frac{3.0 \times 10^{-12}}{80 \times 10^{-6}} = 3.75 \times 10^{-8} \text{ W m}^{-2}$$

$$\beta = 10 \log \frac{3.75 \times 10^{-8}}{10^{-12}} = 46 \text{ dB}$$

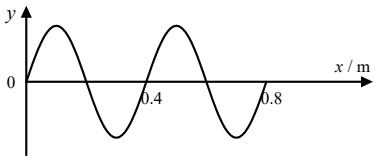
Explanation: 80 mm^2 = 80 × 10^{-6} m^2. Intensity is power per unit area.

5. A signal generator produces sound of frequency 1700 Hz. If the speed of sound in air is 340 m s^{-1}, which graph represents the variation of displacement, y, of the air particles with distance, x, for the sound waves when the signal generator produces the sound?

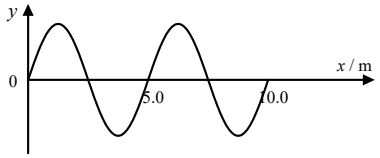
A



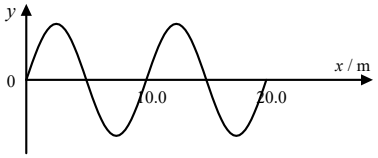
B



C



D



Answer: A

$$\lambda = \frac{v}{f} = \frac{340}{1700} = 0.20 \text{ m}$$

Graph A shows one complete wave every 0.2 m.

Explanation: Read the wavelength from the graph as the distance for one complete cycle.

6. A car driver hears the sound of siren from an ambulance while the car and the ambulance are moving along a straight road. The apparent frequency heard by the car driver is lower than the frequency of sound of the siren, and as the car driver increases the speed of the car, the apparent frequency heard by the car driver increases. Which statement is true about the above scenario?

A The car leaves the ambulance in the same direction.
B The car leaves the ambulance in the opposite direction.
C The car approaches the ambulance in the same direction.
D The car approaches the ambulance in the opposite direction.

Answer: C

The frequency is lower, so the separation is increasing: the ambulance moves away from the car. When the car speeds up the frequency rises, so the car (observer) is moving towards the ambulance. Both move the same way, with the ambulance ahead and faster: the car approaches the ambulance in the same direction.

Explanation: $f' = f\frac{v + v_o}{v + v_s}$: with the source receding faster than the observer follows, $f' < f$, and increasing v_o raises f'.

7. A dentist uses a mirror to observe a patient's tooth. An upright image twice the size of the tooth is observed when the mirror is placed 2.0 cm from the patient's tooth. What is the radius of curvature of the mirror?

A 8.0 cm B 4.0 cm C 2.7 cm D 2.0 cm

Answer: A

Upright, magnified image: virtual, $m = 2 \Longrightarrow v = -4.0$ cm with $u = 2.0$ cm

$$\frac{1}{f} = \frac{1}{2.0} - \frac{1}{4.0} \Longrightarrow f = 4.0 \text{ cm}, r = 8.0 \text{ cm}$$

Explanation: A dentist's mirror is concave, with the tooth inside the focal length.

8. A biconvex thin glass lens with a refractive index of 1.5 has a radius of curvature 10 cm. What are the characteristics of the image formed if an object is placed 5.0 cm from the lens?

A Real, inverted and magnified
B Real, inverted and diminished
C Virtual, upright and magnified
D Virtual, upright and diminished

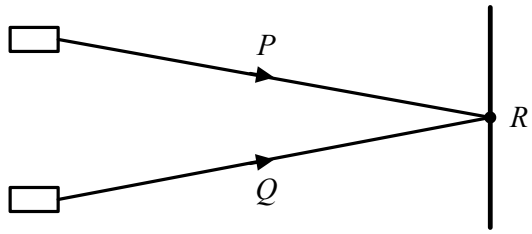
Answer: C

$$\frac{1}{f} = (1.5 - 1)(\frac{1}{10} + \frac{1}{10}) \Longrightarrow f = 10 \text{ cm}$$

The object (5.0 cm) is inside the focal length: virtual, upright and magnified image.

Explanation: A converging lens acts as a magnifying glass when the object is closer than the focal point.

9. Two coherent light rays, P and Q, propagate towards a point, R, as shown in the diagram below.



Which condition produces a maximum constructive interference at R?

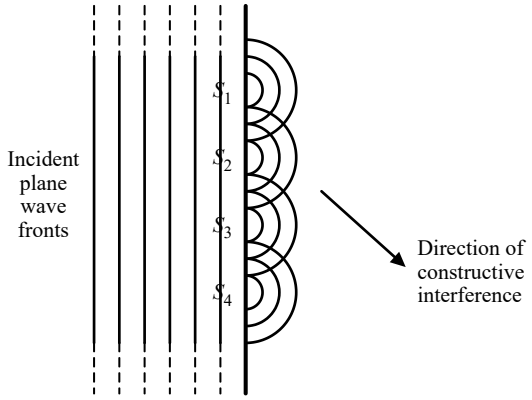
A P and Q are emitted in antiphase and the phase difference at R is 2π.
B P and Q are emitted in antiphase and the path difference at R is λ.
C P and Q are emitted in phase and the phase difference at R is π.
D P and Q are emitted in phase and the path difference at R is λ.

Answer: D

If the sources are in phase, a path difference of a whole number of wavelengths (e.g. λ) gives waves in phase at R: constructive interference.

Explanation: If the sources were in antiphase, a path difference of λ would give destructive interference. A phase difference of π at R is always destructive.

10. In a ripple tank experiment, wave fronts due to diffraction from four slits, S_1, S_2, S_3 and S_4, are shown in the diagram below.



The arrow in the diagram represents one of the directions of the constructive interference. If the wavelength of the waves is λ, what is the path difference between the waves from two consecutive slits?

A λ B 2λ C 3λ D 4λ

Answer: B

Along the arrow, the wavefronts that meet from consecutive slits differ by two whole wavelengths: path difference = 2λ.

Explanation: Constructive interference occurs where the path difference between adjacent slits is a whole number of wavelengths; count the wavefronts between neighbouring slits along the arrow.

11. In a photoelectric experiment, a monochromatic light with certain frequency and intensity is used to emit photoelectrons. Which statement is true about the changes of the photoelectric effect that occurs if the frequency and the intensity are increased?

A The higher intensity produces higher current.
B The higher frequency causes more photoelectrons to be emitted.
C The higher frequency does not change the maximum speed of photoelectrons.
D The higher intensity causes the maximum kinetic energy of photoelectrons to be higher.

Answer: A

A higher intensity means more photons per second, so more photoelectrons are emitted per second: a higher photocurrent.

Explanation: Frequency controls the maximum kinetic energy (and speed) of photoelectrons; intensity controls their number.

12. Electron microscopes show greater magnification than optical microscopes because

A visible light can be diffracted
B electrons travel slower than light
C visible light gives better resolution
D electrons have shorter particle wavelength

Answer: D

Fast electrons have a much shorter de Broglie wavelength than visible light, so they are diffracted less and give better resolution (more useful magnification).

Explanation: Resolution is limited by diffraction, which depends on wavelength.

13. Which statement is **not** true about the line spectra for hydrogen atom?

A An absorption spectrum consists of dark lines.
B The emission lines are due to transitions of electrons from high energy to lower energy levels.
C The absorption lines are due to transitions of electrons from low energy to higher energy levels.
D An emission spectrum is produced when visible light passes through a tube of hydrogen gas.

Answer: D

An emission spectrum is produced by hot, excited hydrogen gas emitting light, not by passing light through the gas (that gives an absorption spectrum). Statement D is not true.

Explanation: Emission lines: electrons drop to lower levels. Absorption (dark) lines: electrons absorb photons and jump to higher levels.

14. Three quarter of a given radioactive source decays in a period of three days. What fraction of the radioactive source remains after six days?

A $\frac{1}{16}$ B $\frac{1}{8}$ C $\frac{3}{4}$ D $\frac{15}{16}$

Answer: A

Three quarters decay in 3 days, so ¼ remains: 3 days = 2 half-lives (half-life 1.5 days).

After 6 days (4 half-lives): $(\frac{1}{2})^4 = \frac{1}{16}$

Explanation: Alternatively, the remaining fraction is multiplied by ¼ every 3 days: ¼ × ¼ = $\frac{1}{16}$.

15. In a nuclear fission reaction, the total mass before fission is 236.053 u and the total mass after fission is 235.836 u. What is the energy released in the reaction?

A 3.92×10^{-25} J B 3.24×10^{-11} J C 2.02×10^{8} J D 1.95×10^{16} J

Answer: B

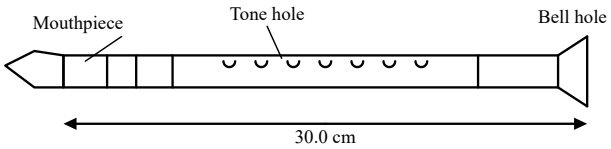
$$\Delta m = 236.053 - 235.836 = 0.217 \text{ u}$$

$$E = (0.217)(1.66 \times 10^{-27})(3.00 \times 10^{8})^2 = 3.24 \times 10^{-11} \text{ J}$$

Explanation: This is about 202 MeV, the typical energy released in one fission of uranium-235.

Section B [15 marks]

16. A recorder with eight tone holes is shown in the diagram below.



Assume that the recorder is a long pipe 30.0 cm with two open ends.
[The speed of sound in air is 330 m s^{-1}.]

(a) Calculate the fundamental resonance frequency of the recorder when all the tone holes are covered. [3]

Answer

Open at both ends: antinodes at both ends, $L = \frac{\lambda}{2}$ — 1

$f = \frac{v}{2L} = \frac{330}{2(0.300)}$ — 1

$f = 550$ Hz — 1

3

Explanation: With all holes covered the air column is the full 30.0 cm length.

(b) Explain the difference between the calculated frequency in (a) and the actual frequency produced. [3]

Answer

The actual frequency is lower than 550 Hz — 1

The antinodes form slightly outside the open ends (end correction), so the effective length of the air column is longer than 30.0 cm — 1

A longer effective length gives a longer wavelength and so a lower frequency (the tube is also not a simple uniform pipe) — 1

3

Explanation: For an open pipe the effective length is about $L + 2(0.6r)$, where r is the radius of the pipe.

(c) If the bell hole is also closed, determine the new fundamental resonance frequency of the recorder. [2]

Answer

Closed at one end: $f = \frac{v}{4L} = \frac{330}{4(0.300)}$ 1

$f = 275$ Hz 1

2

Total for question **8**

Explanation: Closing one end makes a node there, so the fundamental wavelength doubles and the frequency halves.

17. A concave mirror has radius of curvature 20.0 cm. A small pyramid block of height 5.50 cm is placed upright 25.0 cm in front of the mirror.

(a) Calculate the distance of the image. [3]

Answer

$f = \frac{r}{2} = 10.0$ cm 1

$\frac{1}{v} = \frac{1}{10.0} - \frac{1}{25.0}$ 1

$v = 16.7$ cm (real, in front of the mirror) 1

3

Explanation: The object is beyond the centre of curvature (20.0 cm), so the image is between *F* and *C*.

(b) Determine the size of the image. [3]

Answer

$m = \frac{v}{u} = \frac{16.7}{25.0} = 0.667$ 1

Image height = (0.667)(5.50) 1

= 3.67 cm 1

3

Explanation: The magnification is less than 1, so the image is smaller than the object.

(c) State **one** characteristic of the image. [1]

Answer

Any one: real / inverted / diminished — 1

1

Total for question **7**

Explanation: An object beyond *C* of a concave mirror gives a real, inverted, diminished image between *F* and *C*.

Section C [30 marks]

18. (a) State **two** conditions for the formation of standing waves. [2]

Answer

Two waves of the same type, frequency and amplitude (same speed) — 1

travelling in opposite directions superpose (e.g. an incident wave and its reflection) — 1

2

Explanation: The superposition produces fixed nodes and antinodes.

(b) A standing wave on a stretched string is represented by $y = 0.15 \cos (5.0x) \sin (300t)$, where y and x are in metres, and t is in seconds.

(i) Calculate the speed of the standing wave. [4]

Answer

Compare with $y = 2A \cos kx \sin \omega t$: $\omega = 300$ rad s^{-1} — 1

$k = 5.0$ m^{-1} — 1

$v = \frac{\omega}{k}$ — 1

$v = \frac{300}{5.0} = 60$ m s^{-1} — 1

4

Explanation: This is the speed of the two progressive waves that make up the standing wave.

(ii) Determine the distance between two consecutive nodes. [2]

Answer

Distance between consecutive nodes $= \frac{\lambda}{2} = \frac{\pi}{k}$ — 1

$= \frac{\pi}{5.0} = 0.628$ m — 1

2

Explanation: $\lambda = 2\pi/k = 1.26$ m.

(iii) State the position of the string that can be touched without disturbing the standing wave. Give reason to your answer. [2]

Answer

At the nodes, where $\cos(5.0x) = 0$: $x = 0.314$ m, 0.942 m, … 1

because the nodes are always at rest (zero amplitude), so touching them does not change the vibration 1

2

Explanation: Touching an antinode would damp the vibration.

(c) A siren on a train emits sound at frequency 500 Hz while moving towards a station at a speed of 5.0 m s^{-1}. An identical stationary siren is sounded at the station.
[The speed of sound in air is 343 m s^{-1}.]

(i) Determine the beat frequency heard by a stationary observer at the station. [3]

Answer

Train siren heard at the station: $f' = f\frac{v}{v - v_s} = 500\frac{343}{343 - 5.0}$ 1

$f' = 507.4$ Hz 1

Beat frequency = 507.4 − 500 = 7.4 Hz 1

3

Explanation: The beat frequency is the difference between the two frequencies heard.

(ii) Explain the change in beats heard by the observer if the wind blows in the same direction as the train. [2]

Answer

The wind increases the speed of sound from the train towards the station ($v + w$) 1

The apparent frequency $500\frac{v + w}{v + w - 5.0}$ decreases towards 500 Hz, so the beat frequency decreases (fewer beats per second) 1

2

Total for question **15**

Explanation: The station siren is stationary relative to the observer, so the wind does not change its frequency.

19. (a) State Huygens's principle. [3]

Answer

Every point on a wavefront acts as a source of secondary wavelets	1
The secondary wavelets spread forward with the same speed (and frequency) as the original wave	1
The new wavefront is the common tangent (envelope) to the secondary wavelets	1
	3

Explanation: This construction explains reflection, refraction and diffraction.

(b) A hydrogen lamp emits light with two wavelengths, 656 nm and 434 nm. The light is incident normally onto a diffraction grating with 4000 lines per cm and produce diffraction lines on a screen. Determine the angular separation between the two lines in the second order of the diffraction. [4]

Answer

$d = \dfrac{1}{4000 \times 10^2} = 2.50 \times 10^{-6}$ m	1
656 nm: $\sin\theta_1 = \dfrac{2(656 \times 10^{-9})}{2.50 \times 10^{-6}} \Rightarrow \theta_1 = 31.6°$	1
434 nm: $\sin\theta_2 = \dfrac{2(434 \times 10^{-9})}{2.50 \times 10^{-6}} \Rightarrow \theta_2 = 20.3°$	1
Angular separation = 31.6° − 20.3° = 11.3°	1
	4

Explanation: 4000 lines per cm = 4.00×10^5 lines per metre. Use $n = 2$ for both lines.

(c) A monochromatic light is incident into two slits of distance 0.020 cm. An interference pattern is formed on a screen 2.0 m away from the slits. The first bright fringe is 0.35 cm from the centre of the interference pattern.

(i) Determine the wavelength of the monochromatic light. [2]

Answer

$\lambda = \dfrac{yd}{D} = \dfrac{(0.35 \times 10^{-2})(0.020 \times 10^{-2})}{2.0}$	1
$\lambda = 3.5 \times 10^{-7}$ m	1
	2

Explanation: The first bright fringe is one fringe separation from the centre.

(ii) Calculate the distance between the centre of the interference pattern and the third dark fringe. [2]

Answer

Third dark fringe at $2.5y = 2.5(0.35\text{ cm})$ — 1

$= 0.875\text{ cm} = 8.75 \times 10^{-3}\text{ m}$ — 1

2

Explanation: Dark fringes lie halfway between bright fringes: at $0.5y$, $1.5y$, $2.5y$, …

(iii) Explain the change on the interference pattern if the experiment is carried out in water. [4]

Answer

Light travels more slowly in water, with the same frequency — 1

so its wavelength decreases: $\lambda_w = \frac{\lambda}{n}$ — 1

Fringe separation $y = \frac{\lambda D}{d}$ decreases — 1

The fringes become closer together (by a factor of about 1.33) — 1

4

Total for question **15**

Explanation: The positions of the bright and dark fringes depend on the wavelength in the medium between the slits and the screen.

20. (a)

(i) Define *binding energy* of a nucleus. [1]

Answer

The energy required to separate a nucleus completely into its individual protons and neutrons (equal to the energy released when it is formed) — 1

1

Explanation: $E_B = \Delta mc^2$, where Δm is the mass defect.

(ii) Write an equation when $^{238}_{92}\text{U}$ nucleus decays into $^{234}_{90}\text{Th}$. [1]

Answer

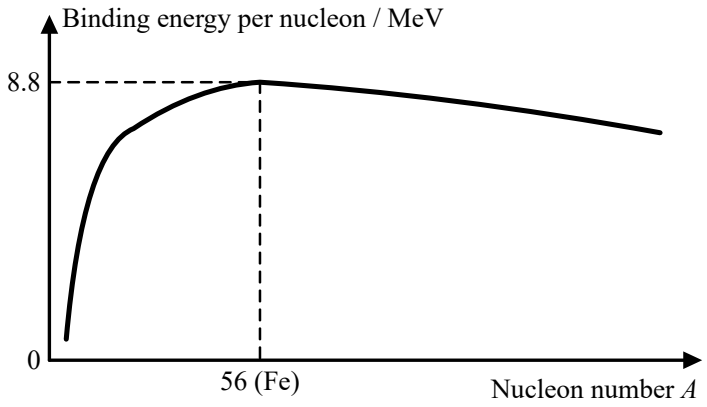
$^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He}$ (+ energy) — 1

1

Explanation: Alpha decay: nucleon number falls by 4, proton number by 2.

(iii) Sketch a graph that shows the variation of binding energy per nucleon against nucleon number. [1]

Answer



Rises to a maximum near $A = 56$ (about 8.8 MeV) then decreases slowly — 1

1

Explanation: Nuclei near iron are the most stable.

(iv) Explain the process in (a)(ii) that can occur spontaneously. [2]

Answer

U-238 is beyond the peak: the products (Th-234 and He-4) have a higher total binding energy (higher binding energy per nucleon) than U-238 — 1

so the total mass of the products is less than that of U-238 and energy is released; the process is energetically favourable and occurs spontaneously — 1

2

Explanation: Any decay that releases energy (positive Q value) can happen spontaneously.

(b) The atomic mass of $^{20}_{10}$Ne is 19.997939 u.

[The mass of electron is 0.000550 u, mass of proton is 1.007825 u, and mass of neutron is 1.008665 u, 1 u ≡ 931.5 MeV.]

(i) Calculate the mass of a $^{20}_{10}$Ne nucleus. [2]

Answer

Nuclear mass = atomic mass − mass of 10 electrons = 19.997939 − 10(0.000550) — 1

= 19.992439 u — 1

2

Explanation: Neon has 10 electrons, which are included in the atomic mass.

(ii) Calculate the total mass of the individual nucleon in the nucleus. [2]

Answer

Total = 10(1.007825) + 10(1.008665) 1

= 20.164900 u 1

2

Explanation: Neon-20 has 10 protons and 10 neutrons.

(iii) Explain the differences between the values in (b)(i) and (b)(ii). [3]

Answer

The total mass of the separate nucleons is greater than the mass of the nucleus 1

The difference is the mass defect, $\Delta m = 0.172461$ u 1

When the nucleons combine to form the nucleus, this mass is converted to energy (the binding energy), $E = \Delta mc^2$ 1

3

Explanation: Energy must be supplied to pull the nucleus apart, which reappears as the extra mass of the separated nucleons.

(iv) Determine the binding energy per nucleon of the nucleus. [3]

Answer

$\Delta m = 20.164900 - 19.992439 = 0.172461$ u 1

$E_B = (0.172461)(931.5) = 160.6$ MeV 1

Per nucleon = $\frac{160.6}{20} = 8.032$ MeV 1

3

Total for question **15**

Explanation: Use 1 u ≡ 931.5 MeV, as given in the question.